



Gopal's Garden High School

Class: 9
Date: 20/12/23

Periodic Test- 3 (2023-24)
Mathematics (Standard)

Marks: 50
Duration: 2h

General Instructions:

1. This Question Paper has 5 Sections A, B, C, D and E.
2. Section A has 13 MCQs carrying 1 mark each
3. Section B has 3 questions carrying 02 marks each.
4. Section C has 4 questions carrying 03 marks each.
5. Section D has 3 questions carrying 05 marks each.
6. Section E has 1 case based integrated units of assessment (04 marks each) with sub- parts of the values of 1, 1 and 2 marks each respectively.
7. All Questions are compulsory.
8. Draw neat figures wherever required. Take $\pi = 22/7$ wherever required if not stated.

Section A

Section A consists of 13 questions of 1 mark each

- 1 The value of $2.\overline{36} + 0.\overline{23}$, when expressed in the simplest form is 1
(a) $\frac{257}{99}$ (b) $\frac{238}{99}$ (c) $\frac{247}{99}$ (d) $\frac{275}{99}$
- 2 If $x^{140} + 2x^{151} + k$ is divisible by $x + 1$, then the value of k is 1
(a) 1 (b) -3 (c) 2 (d) -2
- 3 What do you call a figure formed by two straight lines having a common point? 1
(a) Angle
(b) Triangle
(c) Rhombus
(d) Kite
- 4 Two complementary angles are such that two time the measure of one is equal to three times the measure of the other. The measure of the smaller angle is 1
(a) 45° (b) 30° (c) 36° (d) 28°

- 5 Point $(0, -7)$ lies 1
- (a) on the x-axis
(b) in the fourth quadrant
(c) on the y-axis
(d) in the second quadrant
- 6 The cost of a pen(x) is ten more than thrice the cost of book(y). It can be expressed in the form of linear equation as 1
- (a) $x+3y-10=0$
(b) $x-3y-10=0$
(c) $x+3y+10=0$
(d) $x-10(3y)=0$
- 7 The bisectors of exterior angles at B and C of $\triangle ABC$ meet at O. If $\angle A = x^\circ$, then $\angle BOC =$ 1
- (a) $90^\circ + \frac{x}{2}$ (b) $90^\circ - \frac{x}{2}$ (c) $180^\circ - \frac{x}{2}$ (d) $180^\circ + \frac{x}{2}$
- 8 The bisectors of any two adjacent angles of a parallelogram intersect at 1
- (a) 30° (b) 45° (c) 60° (d) 90°
- 9 The area of a triangle is 150 cm^2 and its sides are in the ratio 3 : 4 : 5. What is its perimeter? 1
- 10 If the mean of x, y and z is k, then the mean of $\frac{x}{y}, 1, \frac{z}{y}$ is 1
- (a) k (b) $\frac{k}{y}$ (c) $\frac{k}{x}$ (d) $\frac{k}{z}$
- 11 One of the zeroes of the polynomial $f(x) = 2x^2 + 7x - 4$ is 1
- (a) 2
(b) $\frac{1}{2}$
(c) $-\frac{1}{2}$
(d) -2

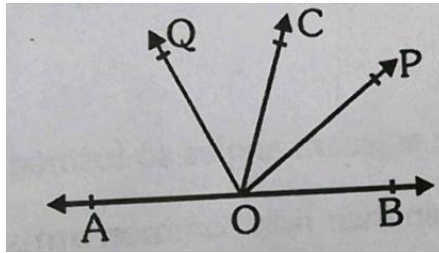
In each of the following questions, a statement of Assertion is given, and corresponding statement of Reason is given just below it. Of the statements, given below, mark the correct answer as:

- (a) Both assertion and reason are true, and reason is the correct explanation of assertion.
(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.
(c) Assertion is true, but reason is false.
(d) Assertion is false, but reason is true.

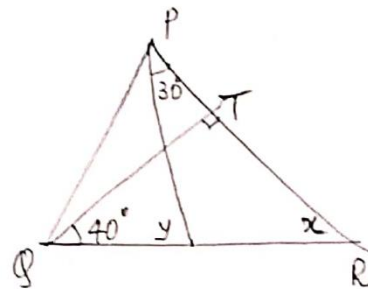
- 12 Statement – I (Assertion): If mean = 24, Median = 26, then mode=30 1
Statement – 2 (Reason): $3 \text{ mean} = 2 \text{ Median} - \text{Mode}$.
- 13 Statement – I (Assertion): The perpendicular distance of a point 1
from y-axis is called its x-coordinate.
Statement – 2 (Reason): The x co-ordinate of the point on y-axis is zero.

Section B
Section B consists of 3 questions of 2 marks each

- 14 AOB is a line. OP bisects $\angle BOC$ and OQ bisects $\angle AOC$, find 2
 $\angle POQ$.

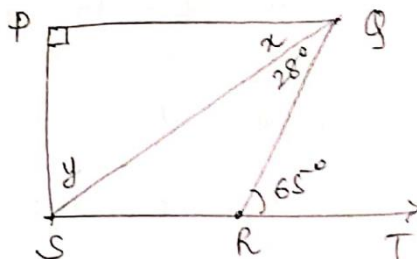


- 15 In the adjoining figure, find x and y. 2



OR

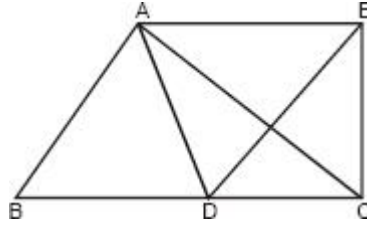
In the adjoining figure,
 $PQ \parallel ST$, find x and y.



- 16 Plot the following points and write the name of the figure obtained 2
by joining, them in order $P(-3, 2)$, $Q(-7, -3)$, $R(6, -3)$ and $S(2, 2)$.

Section C
Section C consists of 4 questions of 3 marks each

- 17 In the given figure, $AB = AD$, $AC = AE$ and $\angle BAD = \angle EAC$, then prove that $BC = DE$. 3



- 18 Prove that 3

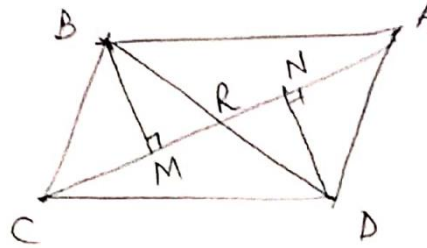
$$\frac{1}{1+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \frac{1}{\sqrt{4}+\sqrt{5}} + \frac{1}{\sqrt{5}+\sqrt{6}} + \frac{1}{\sqrt{6}+\sqrt{7}} + \frac{1}{\sqrt{7}+\sqrt{8}} + \frac{1}{\sqrt{8}+\sqrt{9}} = 2$$

OR

If $x^3 = 28$, $y^3 = 27$, find the value of $x + y - \frac{1}{x^2 + xy + y^2}$.

- 19 P is a point on the bisector of $\angle ABC$. If the line through P, parallel to BA meets BC at Q, prove that BPQ is an isosceles triangle. 3

- 20 In the adjoin figure, $BM = DN$. Prove that AC bisects BD. 3



Section D
Section C consists of 3 questions of 5 marks each

- 21 State and prove the midpoint theorem. Hence show that, in a $\triangle ABC$, if AD and BE are medians, $BE \parallel DF$. Prove that: $CF = \frac{1}{4}AC$. 5

- 22 a) Factorize: $-(a - 3b)^3 + (3b - c)^3 + (c - a)^3$ 5
 b) Without any actual division, prove that the following $2x^4 - 5x^3 + 2x^2 - x + 2$ is divisible by $x^2 - 3x + 2$.

OR

- a) If the polynomials $az^3 + 4z^2 + 3z - 4$ and $z^3 - 4z + a$ leave the same remainder when divided by $z - 3$, find the value of a.
 b) Compute the perimeter of a rectangle whose area is $25x^2 - 35x + 12$.

- 23 (i) The results of pass percentages of class IX and X in a CBSE school for 5 years is given below. 5

Year	2018	2019	2020	2021	2022
Std IX	90	95	90	80	98
Std X	95	80	85	90	95

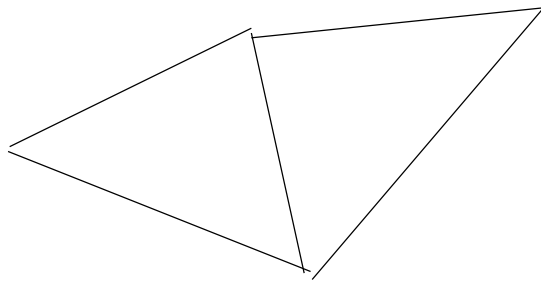
Draw joint bar diagram to represent the data.

(ii) The mean of 200 items was 50. Later on, it was discovered that the two items were misread as 92 and 8 instead of 192 and 88. Find the correct mean.

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Section E Case-Based Question

Sarla Devi has a triangular field with sides 240 m, 200m and 360 m where she grew wheat. In another triangular field, she wanted to grow potatoes in one part and beetroot in another part. He divided the other triangular field into two parts by joining the mid-point of the longest side to the opposite vertex as shown in the figure.



- What is the area of the wheat field? 1
- What is the area of the field used for growing beetroot? Find the ratio of the areas of the field used for growing potatoes and beetroot? 2
- What is the total area of the land she has? 1
